

COCONGRUENCES ON GROUPS ARE EFFECTIVE

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Our goal is to prove that every cocongruence in the category **Grp** is effective. We will establish a more general result for categories in which pushouts and monomorphisms interact in a suitable way.

Definition 1. We shall say that a category $\mathcal{C}$ has *good pushouts of monomorphisms* if it has pushouts of monomorphisms and if, for every diagram of monomorphisms

$$\begin{array}{ccc} B & \longrightarrow & B' \\ \uparrow & & \uparrow \\ A & \longrightarrow & A' \\ \downarrow & & \downarrow \\ C & \longrightarrow & C' \end{array}$$

in which each square is a pullback, the induced morphism

$$B \sqcup_A C \rightarrow B' \sqcup_{A'} C'$$

is also a monomorphism.

Proposition 2. *The category **Grp** has good pushouts of monomorphisms.*

Proof. Consider a diagram as above. We regard every monomorphism in it as an inclusion. Choose a system of representatives $S \subseteq B$ for the right A -cosets in B , meaning that the multiplication map $\cdot : A \times S \rightarrow B$ is bijective. Likewise, choose $T \subseteq C$ such that the multiplication map $\cdot : A \times T \rightarrow C$ is bijective. We may assume that $1 \in S$ and $1 \in T$.

It is well known (see, for example, Serre's book *Trees*, Ch. I, §1, Thm. 1) that every element of the amalgamated free product $B \sqcup_A C$ has a unique representation of the form

$$w = a \cdot x_1 \cdots x_n,$$

where $a \in A$, each x_i lies either in $S \setminus \{1\}$ or in $T \setminus \{1\}$, and these choices alternate.

The map

$$A \backslash B \rightarrow A' \backslash B', \quad Ab \mapsto A'b$$

is injective. Indeed, if $b_1, b_2 \in B$ satisfy $A'b_1 = A'b_2$, then $b_1 b_2^{-1} \in A'$. Since $B \cap A' = A$, it follows that $b_1 b_2^{-1} \in A$, and hence $Ab_1 = Ab_2$.

Therefore, we may extend S to a system of representatives $S' \subseteq B'$ for the right A' -cosets in B' . Likewise, we may extend T to a system of representatives $T' \subseteq C'$ for the right A' -cosets in C' .

With respect to these systems, an element $w \in B \sqcup_A C$ written in normal form as above remains in normal form after being mapped to $B' \sqcup_{A'} C'$. This shows that the induced map is injective. $\square$

Proposition 3. *Let $\mathcal{C}$ be a balanced category with good pushouts of monomorphisms and equalizers of monomorphisms. Then every cocongruence in $\mathcal{C}$ is effective.*

Proof. Let $X \in \mathcal{C}$ be an object, and let $i_1, i_2 : X \rightrightarrows Y$ be a cocongruence. Since it is coreflexive, there exists a morphism $r : Y \rightarrow X$ satisfying

$$r \circ i_1 = \text{id}_X, \quad r \circ i_2 = \text{id}_X.$$

In particular, i_1 and i_2 are monomorphisms. Since the cocongruence is cotransitive, there exists a morphism

$$c : Y \rightarrow Y \sqcup_{i_2, X, i_1} Y$$

satisfying

$$c \circ i_1 = u_1 \circ i_1, \quad c \circ i_2 = u_2 \circ i_2,$$

where $u_1, u_2 : Y \rightrightarrows Y \sqcup_{i_2, X, i_1} Y$ are the pushout inclusions satisfying $u_1 i_2 = u_2 i_1$. We will not use the fact that the cocongruence is cosymmetric; this will follow automatically. Define the monomorphism

$$E := \text{eq}(i_1, i_2) \hookrightarrow X.$$

Since i_1 and i_2 agree on E , there exists a unique morphism

$$\varphi : X \sqcup_E X \rightarrow Y$$

defined by $\varphi \circ j_1 = i_1$ and $\varphi \circ j_2 = i_2$, where $j_1, j_2 : X \rightrightarrows X \sqcup_E X$ are the two inclusions.

We must show that φ is an isomorphism. It is clearly an epimorphism, since i_1 and i_2 are jointly epimorphic by assumption. Since $\mathcal{C}$ is balanced, it therefore suffices to prove that φ is a monomorphism.

We will show that even the morphism

$$\gamma := c \circ \varphi : X \sqcup_E X \rightarrow Y \sqcup_{i_2, X, i_1} Y$$

is a monomorphism. It is characterized by

$$\begin{aligned} \gamma \circ j_1 &= c \circ \varphi \circ j_1 = c \circ i_1 = u_1 \circ i_1, \\ \gamma \circ j_2 &= c \circ \varphi \circ j_2 = c \circ i_2 = u_2 \circ i_2. \end{aligned}$$

In other words, γ is induced by the diagram of monomorphisms

$$\begin{array}{ccc} X & \xrightarrow{i_1} & Y \\ \uparrow & & \uparrow i_2 \\ E & \longrightarrow & X \\ \downarrow & & \downarrow i_1 \\ X & \xrightarrow{i_2} & Y \end{array}$$

Since $\mathcal{C}$ has good pushouts of monomorphisms, it suffices to verify that both squares are pull-backs. Observe that the two squares are symmetric, so it is enough to consider one of them. To verify the universal property, let $a : T \rightarrow X$ and $b : T \rightarrow X$ be morphisms satisfying $i_1 \circ a = i_2 \circ b$. Applying $r : Y \rightarrow X$, we obtain

$$a = r \circ i_1 \circ a = r \circ i_2 \circ b = b.$$

Thus, a is simply a morphism equalizing i_1 and i_2 , so it factors uniquely through $E \hookrightarrow X$. $\square$

Corollary 4. *Every cocongruence in the category $\mathbf{Grp}$ is effective.*